

Study of Hammer Throw Performance in Relation to Selected Basic and Specific Test Performances in Female Throwers

Dr. Jasmail Singh¹, Ms. Suman², Dr. Simarjeet Singh³

ABSTRACT

The aim of the study was to find out relationship between female Hammer Throw performance and selected basic and specific tests. Twentytwo female Hammer Throwers of national level players were selected as subjects for this study. Bench Press, Power Clean, Over Head Throw, 30 M Standing Start, Standing Triple Jump, Standing Vertical Jump, Full Squat, Dead Lift, Snatch, Front Throw, Standing Penta Jump, 1 Turn 4 Kg Hammer Throw, Full Turn 3 Kg Hammer Throw, Full Turn 2 Kg Hammer Throw, Full Turn 5 Kg Hammer Throw were used to assess the basic and specific fitness level. Trials were conducted to assess the competition performance. Correlation of coefficient was calculated with the help of software. The study shows that the selected basic and specific test items are highly related to Hammer throw performance.

INTRODUCTION

The history shows roots of Hammer throw in Scotland and Ireland. Since 1900, it is a part of Olympic Games in men's category. In the year 1995 it was introduced in women's category too. The men's Hammer weighs 7.26 kg and measures 121.3 cm in length, and the women's Hammer weighs 4 kg and 119.4 cm in length. The difference in implement weight does factor into the amount of force needed to overcome inertia and to balance against the centripetal forces.

Hammer throwing is an exciting, challenging and fitness demanding event in athletics. Efficient technique and other motor qualities like strength, speed (power), co-ordination, flexibility and mobility, suppleness, skill, training, motivation and

emotional level etc. are the factors determining performance of Hammer throwers. Jabs (1965) stated that good timing, coordination and rhythm are prerequisites in Hammer throw. Bush with Weiskopf (1978) linked quickness and explosiveness as qualities of an ideal Hammer thrower.

Johnson (1969) updated that to throw the Hammer a thrower has strength requirements closely similar to those of the power lifter.

The Hammer thrower needs strength of hamstrings, thighs, and hips. Body mass and strength of the legs, trunk and arms are the most important contributors to increase the speed of the Hammer at release (Ecker, 1974). The athlete delivers the Hammer by making an explosive movement with the

1. Scientific Officer, SAI, NSNIS, Patiala.

2. Athletics Coach, Haryana

3. Scientific Officer (Retd.), SAI, NSNIS, Patiala.

legs, arms, and trunk coordinated in one action, to send the Hammer as far as possible.

Apart from heavy physical strength, it requires excellent foot work coordination. Considering the importance of technique Hildreth (1962) stated that Hammer men must train regularly for strength with weights and resistance exercises; but, this should not be allowed to supersede training for technique. The technique training and strength should progress together so that the strength gained is realized into the correct technique.

This present study was done to know the relationship of general and specific fitness and technical level with performance in women Hammer throwers of India.

METHODOLOGY

Sample

For the study, a total number of 22 athletes were recruited from the different places of India, attending various National coaching camps both national and international. For the study a total no. of 22 athletes in hammer throw were selected. The data was collected from the athletes while attending national camps in preparation for national and international competition. the sample population were from different parts of the country.

Selection of Tests

Six Basic and nine Specific Tests recommended by Poppe (2001) were conducted to collect the data.

Basic Tests

Sr. No.	Test Items	Aim of test
1	30 m Standing Start (sec.)	Speed
2	Standing Vertical Jump (cm)	Leg Explosive Strength
3	Standing Triple Jump (m)	Leg Explosive strength
4	Overhead Shot Throw 4 kg (m)	General Throwing Strength
5	Bench Press (kg)	Maximum Strength
6	Power Clean (kg)	Maximum Strength

Specific Tests

Sr. No.	Test Items	Aim of test
1	Hammer Throw 4 kg one Turn (M)	Special Strength
2	Hammer Throw 3 kg (M)	Special Speed
3	Hammer Throw 2 kg (M)	Special Speed
4	Hammer Throw 5 kg (M)	Special Strength
5	Penta Jump Standing (M)	Leg Explosive Strength
6	Shot 4 Kg Front throw (M)	Explosive Strength
7	Snatch (kg)	Maximum Strength
8	Front Squat (kg)	Maximum Strength
9	Dead Lift (kg)	Maximum Strength

In addition to above given tests, Hammer Throw performance with standard weight was also recorded.

Statistical Procedure

Coefficient of correlation was calculated to find out the relationship between competition performance and selected independent variables.

RESULT & DISCUSSION

Table-1: Mean and SD of selected Basic test items

Test Items	Means	SD
Bench Press	37.16	7.331
Power Clean	60.23	14.758
Over Head Throw	13.3	1.122
30 m Standing Start	5.06	0.279
Standing Triple Jump	5.6	0.354
Standing Vertical Jump	45.73	6.288

Table-2: Mean and SD of selected Specific test items

Test items	Means	Sd
Full Squat	95.05	30.153
Dead Lift	112.27	20.972
Snatch	46.14	7.389
Front Throw	12.17	1.242
Standing Penta Jump	8.05	0.904
1 Turn 4 Kg Hammer Throw	36.04	6.088
Full Turn 3 Kg Hammer Throw	54.16	8.942
Full Turn 2 Kg Hammer Throw	59.18	8.728
Full Turn 5 Kg Hammer Throw	38.78	7.738

Table-3: Relationship between Hammer throw performance and selected basic tests

Test Items	r' value
Standard weight Hammer throw vs bench press	0.784**
Standard weight Hammer throw vs power clean	0.866**
Standard weight Hammer throw vs over head throw	0.864**
30 m standing start	0.718**
Standard weight Hammer throw vs standing triple jump	0.703**
Standard weight Hammer throw vs standing vertical jump	0.811**

**1% level of significance

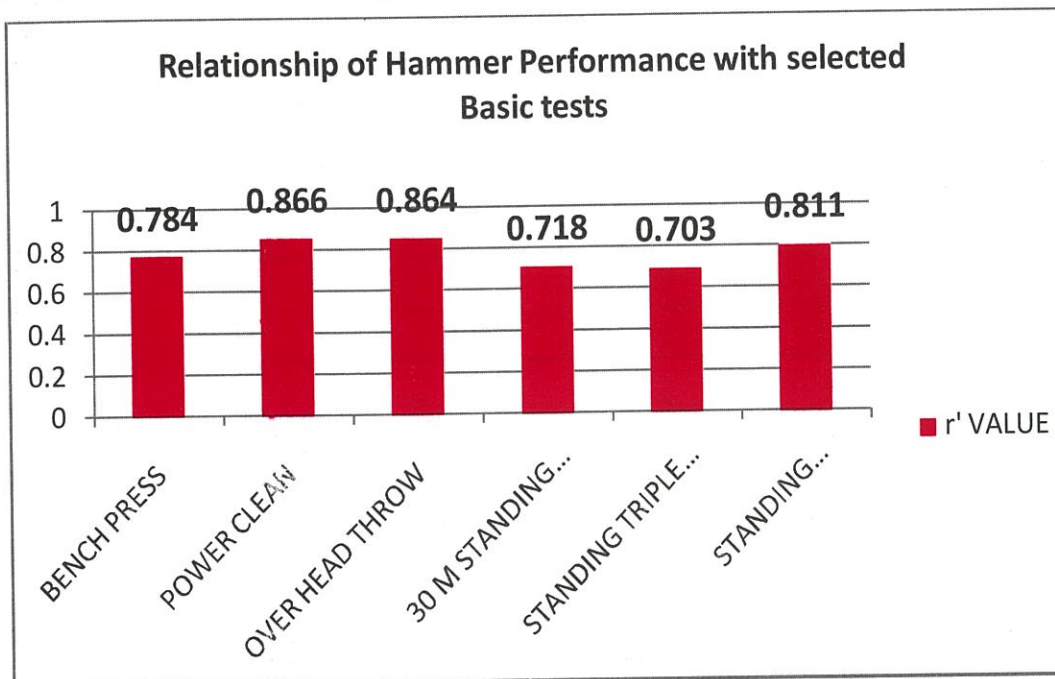


Fig.1: Relationship between Hammer Throw Performance and selected Basic Tests.

Table-4: Relationship between Hammer throw performance and selected specific tests

Test items	r' Value
Standard Weight Hammer Throw Vs Full Squat	0.881**
Standard Weight Hammer Throw Vs Dead Lift	0.928**
Standard Weight Hammer Throw Vs Snatch	0.880**
Standard Weight Hammer Throw Vs Front Throw	0.839**
Standard Weight Hammer Throw Vs standing penta Jump	0.858**
Standard Weight Hammer Throw Vs 1 Turn 4 Kg Hammer Throw	0.874**
Standard Weight Hammer Throw Vs Full turn 3 Kg Hammer Throw	0.958**
Standard Weight Hammer Throw Vs Full Turn 2 Kg Hammer Throw	0.975**
Standard Weight Hammer Throw Vs Full turn 5 Kg Hammer Throw	0.846**

**1% level of significance

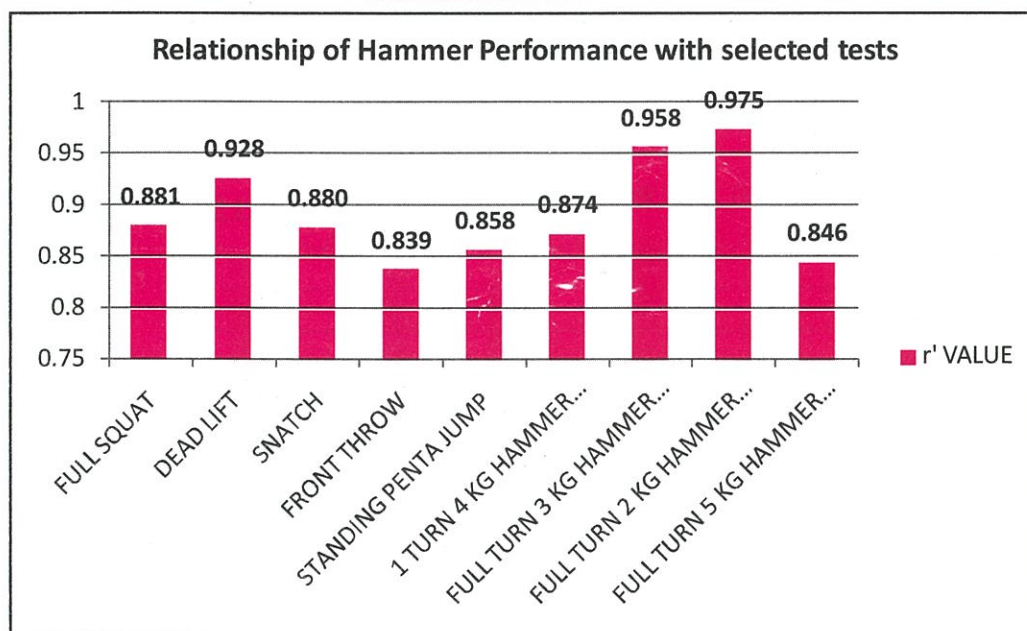


Fig.2: Relationship between Hammer Throw Performance and selected specific Tests.

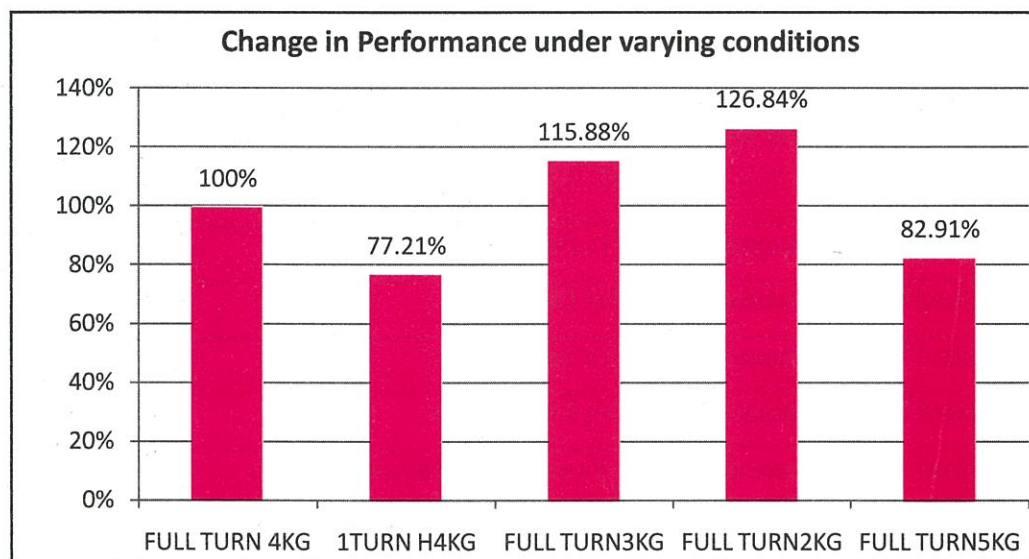


Fig. 3: Change in Hammer Throw Performance under different conditions in %

The results presented in Table 3 show that selected basic tests are highly related to Hammer Throw competition performance. The achieved 'r' values 0.784,

.866, .864, .718, .703; and .811 showed statistically significant relationship of competition performance, with Bench Press, Power Clean, Overhead backward

Throw, 30 m run from standing start, standing triple jump and Standing Vertical Jump, respectively. Standing triple jump showed minimum relationship with performance, with a value of .703; and Power cleans showed the highest 'r' value i.e. .866, from the list of selected basic tests.

The results presented in Table 4 showed a very high relationship between Hammer Throw performance and selected specific tests. Hammer Throw is considered to be highly correlated with maximum strength tests with Full Squat (.881), Dead Lift (.928) and Snatch (.880). Dead Lift stands at number one with the highest 'r' value. Boson (1972) recommended Dead Lift with reverse grip for Hammer Throwers. Vidyasagar (1979) states that maximum strength is essential to meet the demands of strong counter pull of centrifugal force for execution of the technique in Hammer throwing. The achieved significant 'r' values of .839 (Front Throw) and .858 (Standing Penta Jump) show that Explosive strength of shoulder and leg muscles is an important Hammer Throw performance contributor. Full turn 3 kg Hammer Throw and full turn 2 kg Hammer Throw tests considered to be specific speed tests and 1 Turn 4 kg Hammer Throw and Full turn 5 kg Hammer Throw tests are prerequisites for the assessment of competitive performance of Hammer throwers. Judge (2006) stated that throwing

heavy implements is way to build specific strength.

Alford (1958) stated that strength and speed are important qualities in addition to physique, joint mobility, endurance and skill and suggested sprints, uphill shot fast runs, throwing weights, hopping, bounding and jumping exercises.

Full squat, Clean, Snatch, short sprints. Two handed Overhead Backward Throw and forward Throw are some of the suggested exercises for Hammer Thrower's training by Dunn (1976). According to O'Shea (1979) a high level of total strength is essential to achieve high performance in international competitions.

CONCLUSIONS

On the basis of results, it is being concluded that Bench Press, Power Clean, Over Head Throw, 30 M Standing Start, Standing Triple Jump, Standing Vertical Jump, Full Squat, Dead Lift, Snatch, Front Throw, Standing Penta Jump, 1 Turn 4 Kg Hammer Throw, Full Turn 3 Kg Hammer Throw, Full Turn 2 Kg Hammer Throw, Full Turn 5 Kg Hammer Throw are highly related to female Hammer performance. The training to improve the performance in above given tests is being recommended. It is also being concluded that norms suggested by Poppe, D. can be used to analyse and evaluate the Hammer Throw performance and to develop the qualities lacking, to the required level.

REFERENCES

- Alford (1988). NSA Round Table, New Studies in Athletics by IAAF No. 1:p.22.
 Bosen, Ken O.(1972). Training without straining. Delhi: Scholars foundation: p.157.
 Bush, J. & Weiskopf, D. (1978). Track &Field. Boston. Allyn and Bacon: p.241, 281, 297.
 Dunn (1976). Weight training for preparatory throwers. Track Technique No. 65:p.2060.

- Ecker, T.(1974).** Track & Field, USA. Tafnews Press, p.115.
- Hildreth, P.(1962).** Athletics Times Educational Supplement. Great Britain: Taylor Garnett Evens and Co. Ltd. p.155.
- Jabs, Rolfgunter (1965).** Hammer Throwing Principles. Track Technique No. 21: p.670.
- Johnson, C. (1969)** Hammer Throwing London. B.A.A.F.:pp.39-51.
- Judge, Larry (2006).** Rethinking your approach to training for the weight throws. Track Coach, No. 176: pp.5615-5617, 5620.
- O'Shea (1979).** Super quality strength training for the elite athlete, Track & Field Quarterly Review, V.79, No. 4: p.54.
- Poppe, D. (2001).** "Trajectoire" Paris.
- Vidyasagar (1979).** Training systematic in Throwing, NSNIS, Patiala (India): pp. 232-245.